

Mehul Dinesh

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Skills

Languages: Python, JavaScript, SQL, Bash

Frameworks: FastAPI, React 18, Node.js

AI/LLM: RAG Systems, Transformers

Databases: Qdrant, MongoDB, PostgreSQL

Tools/DevOps: Docker (Containerization), GitHub

Actions (CI/CD), Git

Education

PSG College of Technology, Coimbatore

B.E. in Computer Science and Engineering (AI & ML)

Oct 2022 – Present

Experience

Software Engineering Intern, Tech Mahindra, Hyderabad

July 2025 – August 2025

- Designed scalable GCS folder structures to improve multi-format workflow data ingestion and retrieval for video pipelines, enhancing discoverability.
- Automated legacy data restructuring with Python, enabling dynamic schema support and reducing manual pre-processing.

Projects

Veronica — Personal AI Assistant & Multimedia Orchestrator ([GitHub](#))

- Optimized system responsiveness by 30% by architecting an asynchronous multi-threading layer that resolved concurrency bottlenecks between Whisper STT and real-time intent extraction .
- Engineered a unified AI orchestration system integrating 4+ high-performance APIs (Spotify, Gemini 2.0, ElevenLabs, Tavily) into a production-grade FastAPI/React framework with persistent RAG memory.
- Achieved 100% environment parity for a 10GB+ multi-modal project by containerizing dense AI dependencies and automating delivery via a custom GitHub Actions CI/CD pipeline .

Kaizen — AI-Powered Assistant (RAG System) ([GitHub](#))

- Built a production-grade RAG assistant, improving answer accuracy by 40%.
- Developed FastAPI backend serving a fine-tuned Transformer for intent classification (92% test accuracy).
- Engineered LangChain + Qdrant pipeline to generate knowledge-grounded responses from uploaded documents.

PixelPerfect — Hackathon Winner (3rd Place) ([GitHub](#))

- Developed FastAPI backend handling 1,000 requests/minute for image generation and metadata management.
- Integrated Stable Diffusion and ESRRGAN for high-resolution outputs; Gemini vector search accelerated retrieval by 5×.

Publications

MINT: Meta-Learning for Intelligent Prompt Creation, Proceedings of *NMITCON 2025*

- Co-authored work on adaptive prompt generation using hierarchical context encoding and RLHF, improving personalization and diversity.